Do in-silico metrics filter designed enzymes for catalysis?

Supplementary Information

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1 About these tables

This document describes the 31 supplementary tables (S1 to S31) and the six data tables of the Results (T5 to T10) of the paper. It does not reprint them. Every one of these tables is released as a CSV file (Tables 1 to 4, in the Methods, are descriptive and are not released as files); each supplementary entry below gives the path, the number of rows and columns, what a row is, what the key columns mean, an excerpt of the first rows, the places in the paper that cite the table, and caveats; the data tables of the Results are listed with their columns in Appendix A. Supplementary tables are numbered in the order in which the Methods and Results first cite them.

```
README.md          what is released, what was changed, licence
INDEX.csv          one row per table: id, file, title, use, rows, columns, sha256
main/              T5 ... T10    Tables 5 to 10 of the paper
supplementary/     S1 ... S31    supplementary tables, one CSV per table
```

How to read an entry. *File* is the path of the CSV, its size and its licence flag. *Supports* is the part of the paper the table backs. *Cited in* lists the sections of the Methods and Results that cite it (a link opens the paper at that section). *A row is* says what one row represents. *Key columns* explains the columns that matter; the shared provenance block (below) is not repeated. The *excerpt* shows real rows from the file.

2 Conventions shared by the tables

Shared provenance columns. The result tables (T5 to T10 and the supplementary tables whose entry says so) carry the same block of up to 11 columns, so that a row can be traced to its denominator: `claim_type`, `unit` and `n_units` (what was counted and how many), `n_clusters` and `cluster_def` (the resampling unit: enzyme sub-subclass for natural enzymes, campaign and plate well or backbone cluster for designs, position for BglB), `k`, `N` and `k_of_N` (a count with its denominator), `denominator_def`, `status` and `licence_flag`.

Claim types. `specificity` (catalytic against matched control), `equivalence` (panel ratio against 1), `discrimination` (a real contrast, dead against active), `ranking` (against a measured label), `enrichment`.

Applicability status (in the `applicability_status` columns). `OK` can be tested; `OK-KNOCKON` tested but reported as displacement of the other catalytic residues, with no specificity verdict; `OK-PROVISIONAL` the raw status of the interface terms in the re-predicted ladder, resolved by testability; `GLOBAL` tested but has no site input (the PLACER ensemble summaries); `X-BOOKKEEP` a counter or constant; `X-NOINPUT` the input cannot reach the changed quantity; `X-ARM` the metric belongs to the other kind (structure space or prediction) than the test; `X-UNDEF-COV` defined on too few units; `X-CONFOUND` the control is confounded; `NOT-RUN` the metric was not computed for that analysis.

Verdicts in the specificity tables. `specific` (response significant, interval of the specificity ratio above 1, control adequate); `non_specific` (responds, not more than the control); `blind` (does not respond); `invariant` (does not change); and counts-floor labels for cells with too few units.

Why some metrics are not in the main set. A metric with no residue-set input cannot respond *specifically* to catalytic residues, and a metric whose input cannot reach the changed quantity cannot register the lesion at all; such metrics are kept out of the main set. One further quantity passes these tests but is kept out of the main set: the Chai-1 combined score, which Chai-1 uses to rank its own models and which equals $0.2 \times$ pTM + $0.8 \times$ ipTM $- 100 \times$ (inter-chain clash flag), so that it is a rescaled ipTM and not an independent metric. It is still scored, and its role and reason are in S1.

Names that recur. `canonical_key` is the distinct metric after aliases and the per-model summaries of one score are merged (`metric_name` may be an alias or a summary); `in_main_set` is 1 for the 33 reported metrics and for their aliases and per-model summaries; `role` is `main` or a supplement role (see S1; the data tables of the Results use `main`, `supplementary` and `bookkeeping`); `source` is structure-space, prediction-based, PLACER, natural or denovo as the table says; `lesion_test` is the catalytic lesion or the second-shell lesion; `level` or `lesion_step` is the lesion step (isosteric 15.3, non-isosteric 36.4, Ala 53.2, Gly 80.5 Å^3 median side-chain volume change; `second_shell_1/2/4` is the second-shell dose); `ame_flavour` is `crystal` (against the deposited structure) or `self_design` (against a design's own model) and the two are never pooled; `sr_*` is a specificity ratio with its interval; `auroc` is direction-free unless a column says `signed`, `directed` or `magnitude` (in S9 it is the AUROC of the size of the change and is not direction-free); `higher_in` is the group in which the metric takes larger values (`mixed` or `tie` when neither); `sr_median` of `inf` means that the median of the control arm is zero; `rho` is a Spearman correlation.

Licence. Tables flagged `D2DCure_aggregate` in `INDEX.csv` are built from the D2DCure BglB data, whose licence is unstated; they hold metric-level aggregates only, no per-variant row. All other tables are `public`.

Integrity. `INDEX.csv` gives the sha256 of every released file.

3 Where do I find . . . ?

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Table	Title	Topic	Rows × cols
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S4	Definitions of the pair sets	Activity discrimination (3.1)	6 × 4
S5	Zymogen-mature evaluation set: all 97 metrics ranked by AUROC (complete Table 5)	Activity discrimination (3.1)	97 × 26
S6	All 71 metrics ranked by the AUROC of active against inactive on the 192 designs (complete Table 6)	Activity discrimination (3.1)	71 × 28
S7	Substrate swap on all systems: the prediction-based quantities, natural enzymes and de novo designs	Substrate discrimination (3.2)	138 × 17
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S9	BglB: every analysed quantity ranked by rank correlation with the measured impairment (complete Table 8)	Activity ranking (3.3)	54 × 30
S10	BglB: all 39 structure-space metrics	Activity ranking (3.3)	39 × 26
S11	BglB: prediction-based metrics (Chai-1 with the assay substrate)	Activity ranking (3.3)	15 × 26
S12	Hits per 96-well plate for every metric on the 192 designs (filter-style analysis, active against inactive)	Activity ranking (3.3)	71 × 36
S13	Every metric ranked by its correlation with kcat/KM over the 16 designs that have a value (complete Table 9)	Activity ranking (3.3)	52 × 26
S14	Enrichment on the 192 designs at every keep fraction, structure-space metrics (active against inactive)	Activity ranking (3.3)	440 × 26
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S20	Structure-space specificity by metric family and lesion step, with the panel statistic and sensitivity sets	Detection ability (3.4)	20 × 26
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S24	The specific cells that survive the controls	Detection ability (3.4)	70 × 10
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S26	Panel equivalence statistic for the main-set metrics	Detection ability (3.4)	2 × 10
S27	Re-predicted ladder: specificity ratio per step for all metrics	Detection ability (3.4)	140 × 15
S28	Second-shell lesion: every metric scored, with R at each dose and the per-dose flags	Detection ability (3.4)	37 × 58
S29	Effective dimensionality of the panel	Detection ability (3.4)	2 × 10
S30	De novo designs: burial equivalence of the created-site control	Detection ability (3.4)	4 × 11
S31	PLACER ensemble metrics on the isosteric step (not interpretable)	Detection ability (3.4)	50 × 12

5 Supplementary tables

Table S1. Main metric set: the role of every distinct metric, and why

File. `supplementary/S1_main_metric_set.csv` · 139 rows × 16 columns · public

Supports. Methods, Table 1 and 'Which metrics are reported'; the main set of 33 metrics used in Tables 5 to 10. **Cited in.** Methods 2.1.

A row is one distinct metric (a canonical key; aliases and per-model summaries of one score are merged), 139 in all.

The audit that fixes which metrics the paper reports. For each distinct metric it gives the applicability status in the detection test of the catalytic lesion, in the dead-against-active tests and in the activity-ranking tests, and the resulting role: main (33, valid and tested in the deliberate-change tests and in the tests on real proteins), excluded_bookkeeping (61 counters and constants) or one of the supplement roles (supp_placer 25, supp_excluded 10, supp_aggregate 4, supp_partial 3, supp_real_protein_tests_only 2, and supp_derived 1, the Chai-1 combined score, a rescaled ipTM). The reason for each role is in the last column. The status is decided from what a metric reads, whether it is defined and the design of the experiment, never from a result.

Key columns.

Column	Meaning
`canonical_key` / `members` / `rep_metric`	the distinct metric, the names merged into it and the representative one
`mode` / `input_class` / `subfamily` / `aggregator`	structure-space or prediction-based, what the metric reads, its family and how it is summarised
`status_detection_test` / `fraction_detection_test`	applicability status in the detection test of the catalytic lesion and the share of units on which the metric is defined
`status_dead_vs_active_tests` / `fraction_dead_vs_active_tests` / `partial_dead_vs_active_tests`	the same for the dead-against-active tests
`status_ranking_tests` / `ranking_tests_ok`	status in each activity-ranking test (BglB variants, the 16 designs that have a kcat/KM) and the tests in which the metric is OK
`role`	main, excluded_bookkeeping, supp_placer, supp_excluded, supp_aggregate, supp_partial, supp_real_protein_tests_only or supp_derived
`reason`	why the metric has that role

Excerpt (first 3 rows where role = main).

canonical_key	input_class	status_detection_test	status_dead_vs_active_tests	role
ame_crystal_clash_max	pred_interface	OK-PROVISIONAL	OK	main
ame_crystal_rmsd	ame	OK-KNOCKON	OK	main
catalytic_frac_buried	struct_site	OK	OK	main

Notes.

- The status vocabulary is in the Conventions.
- NOT-RUN in status_ranking_tests marks the PLACER metrics, which were not computed in the BglB and plated-design analyses.

Table S2. Metric contracts: what each named quantity reads, over which residues, and where it is valid

File. `supplementary/S2_metric_contracts.csv` · 155 rows × 19 columns · public

Supports. Methods, Table 1 and 'Which metrics are reported'. **Cited in.** Methods 2.1.

A row is one named quantity (155 in all; an alias is a separate row linked by alias_of and canonical_key).

The input contract of every quantity in the panel: what it reads (site-scoped structure, predictor output, active-site accuracy, PLACER output, deposit metadata or bookkeeping), over which residues it is scored in the catalytic arm and in the two kinds of control arm, which inputs it needs, which kinds of metric it is valid for and which changes it cannot respond to.

Key columns.

Column	Meaning
`metric_name` / `canonical_key` / `alias_of` / `alias_kind`	the quantity, the distinct metric after aliases are merged, and the kind of alias
`aggregator` / `metric_family` / `subfamily`	how it is summarised, and its family and subfamily
`input_class`	site-scoped structure, prediction, active-site accuracy (AME), PLACER, deposit metadata or bookkeeping
`scope_catalytic_arm` / `scope_control_arm_catalytic_lesion` / `scope_control_arm_second_shell`	the residues the metric is scored over in the catalytic arm, in the control arm of the catalytic lesion and in the control arm of the second-shell lesion
`needs_cat_residue` / `needs_ligand` / `needs_reference` / `needs_predictor`	the inputs the metric needs
`valid_modes`	structure-space, prediction-based, PLACER, any or deposit metadata
`invariant_under`	changes the metric cannot register (for example the amino-acid propensity term under a rotamer change)
`ame_flavour` / `note`	for active-site accuracy, the reference it is measured against; remarks on the metric

Excerpt (first 3 rows where input_class = struct_site).

metric_name	input_class	scope_catalytic_arm	valid_modes
catalytic_frac_buried	struct_site	scored_residue_set	structure-space
catalytic_ligand_min_dist	struct_site	scored_residue_set+ligand_within_...	structure-space
catalytic_pairwise_dist_max	struct_site	scored_residue_set	structure-space

Table S3. Wider set of 49 zymogen-mature pairs: all 88 scored metrics, with the same-state null

File. `supplementary/S3_zymogen_all_metrics.csv` · 88 rows × 24 columns · public

Supports. Results 3.1.1 (wider set); Methods 2.2.1. **Cited in.** Methods 2.2.1; Results 3.1.1.

A row is one scored metric (88 in all).

The analysis on the 49 pairs for which structure-space metrics were scored: the direction-free AUROC of dead against active with interval, the directed AUROC, the number of pairs, and the AUROC of the same metric on 96 same-state null pairs, which shows how much separation arises without any dead/active difference.

Key columns.

Column	Meaning
`metric_name` / `canonical_key` / `input_class` / `in_main_set`	the metric, its class and whether it is a main-set metric
`n_pairs`	pairs with a value (49 at most)
`auroc` / `auroc_lo` / `auroc_hi` / `auroc_directed`	direction-free AUROC with 95% interval over pairs, and the directed value

Column	Meaning
auroc_vs_same_state_null / n_zymogen_pairs / n_same_state_null_pairs	AUROC of zymogens against the same-state null pairs and the numbers in each group
applicability_status	applicability status

The shared block of 11 provenance columns (see Conventions) is also present and is not repeated above.

Excerpt (first 3 rows).

metric_name	in_main_set	n_pairs	auroc	auroc_lo	auroc_hi
catalytic_frac_buried	1	49	0.5321	0.5019	0.6112
catalytic_hydration_count	0	48	0.5686	0.505	0.6604
catalytic_ligand_min_dist	0	25	0.5264	0.5024	0.6464

Notes.

- Structure-space and PLACER metrics only; the prediction-based metrics were scored on the 21-pair evaluation set (S5).
- Results are descriptive of a wider set; the evaluation set is the 21 pairs of S5.

Table S4. Definitions of the pair sets

File. supplementary/S4_pair_set_definitions.csv · 6 rows × 4 columns · public

Supports. Methods 2.2.1; Results 3.1.1. **Cited in.** Methods 2.2.1; Results 3.1.1.

A row is one set of zymogen-mature pairs (6 in all).

The six sets used in 3.1 with their size and definition: 55 candidate rows, 49 scored pairs, 45 with PLACER, 28 re-predicted pairs, the 21-pair evaluation set (pairs that share a ligand) and the 96 same-state null pairs, and which table uses each.

Key columns.

Column	Meaning
pair_set / n	the set and its size
definition	how the set is defined
used_for	the table that uses it

Excerpt (first 3 rows).

pair_set	n	used_for
zymogen candidate rows	55	candidate list
scored pairs (wider set)	49	S3 (49 = max n_pairs)
pairs with PLACER	45	S3 (the PLACER rows, n_pairs = 45)

Table S5. Zymogen-mature evaluation set: all 97 metrics ranked by AUROC (complete Table 5)

File. `supplementary/S5_zymogen_21pair_all_metrics.csv` · 97 rows × 26 columns · public

Supports. Results 3.1.1; the complete version of Table 5. **Cited in.** Methods 2.2.1; Results 3.1.1.

A row is one metric scored on the 21 pairs (97 in all).

Every metric scored on the 21 pairs that share a ligand: structure-space, PLACER and prediction-based metrics, ranked by direction-free AUROC of dead against active. The best-of-158 threshold (95th percentile of the best AUROC among 158 unrelated quantities in label-swap simulations) is marked: no metric exceeds it.

Key columns.

Column	Meaning
`rank` / `rank_in_table_5`	rank among all 97 and rank among the 33 items of Table 5
`metric_name` / `canonical_key` / `source` / `input_class`	the metric, the distinct metric, structure-space / PLACER / prediction-based, and class
`role`	main or a supplement role (43 rows carry main because aliases of the 33 main-set metrics are separate rows)
`n_pairs` / `auroc` / `auroc_ci_lo` / `auroc_ci_hi`	pairs with a value (21 for every main-set metric) and AUROC with 95% interval over pairs
`higher_in` / `auroc_directed`	the form (dead or active) in which the metric is larger, and the directed AUROC
`above_best_of_158_threshold` / `applicability_status`	1 if above the threshold; applicability status

The shared block of 11 provenance columns (see Conventions) is also present and is not repeated above.

Excerpt (first 3 rows).

rank	metric_name	role	auroc	auroc_ci_lo	auroc_ci_hi
1	catalytic_hydration_count	supp_real_protein_tests_only	0.655	0.5225	0.7913
2	placer_plddt_pde_best	supp_placer	0.6343	0.5097	0.7784
3	placer_plddt_pde_max	supp_placer	0.6343	0.5097	0.7784

Table S6. All 71 metrics ranked by the AUROC of active against inactive on the 192 designs (complete Table 6)

File. `supplementary/S6_plates_all_metrics_auroc.csv` · 71 rows × 28 columns · public

Supports. Results 3.1.2; the complete version of Table 6. **Cited in.** Methods 2.2.2; Results 3.1.2.

A row is one metric scored on the 192 designs (71 in all).

Every metric scored on the 192 plated designs, ranked by the direction-free AUROC of the 16 active designs against the 176 inactive designs, with 95% interval over the 136 backbone clusters. The best-of-128 threshold (permuting labels among designs) is marked: four metrics exceed it. They are two main-set metrics (the intra-residue repulsion, 0.80, and the repulsion of the catalytic residues, 0.78) and two aliases or per-residue variants of the latter.

Key columns.

Column	Meaning
rank / rank_in_table_6 / in_table_6	rank among 71, rank among the 31 items of Table 6 and membership of Table 6
metric_name / canonical_key / source / input_class / role	the metric, its source, class and role
n_designs / n_active	designs with a value (192; 182 for two metrics) and active designs (16)
auroc / auroc_ci_lo / auroc_ci_hi / auroc_directed / higher_in	AUROC, interval, directed AUROC and the group in which the metric is larger
above_best_of_all_metrics_threshold / applicability_status	1 if above the best-of-128 threshold; applicability status

The shared block of 11 provenance columns (see Conventions) is also present and is not repeated above.

Excerpt (first 3 rows).

rank	metric_name	role	auroc	auroc_ci_lo	auroc_ci_hi
1	rosetta_cat_fa_intra_rep	main	0.8018	0.6385	0.8929
2	rosetta_cat_fa_rep	main	0.7765	0.5714	0.918
3	rosetta_catalytic_vdw_rep	main	0.7765	0.5714	0.918

Notes.

- Designs are not exchangeable (10 of the 16 active designs lie in two backbone clusters), so the threshold understates the chance range.

Table S7. Substrate swap on all systems: the prediction-based quantities, natural enzymes and de novo designs

File. supplementary/S7_substrate_swap_per_metric.csv · 138 rows × 17 columns · public

Supports. Results 3.2; Table 7 on all systems. **Cited in.** Methods 2.3; Results 3.2.

A row is one prediction-based quantity under one wrong-ligand condition, for one source (natural enzymes or de novo designs).

The substrate swap on all 59 natural enzymes (cognate value averaged over five seeds) and all 30 de novo designs, for every prediction-based quantity that was scored, including the Chai-1 combined score (a rescaled ipTM) that Table 7 leaves out. For each quantity and each wrong ligand (same enzyme class, different class, decoy; apo for the quantities that are defined without a ligand) it gives the mean under the cognate and under the other condition, the paired change, the change in units of seed noise, and the direction-free AUROC with its interval.

Key columns.

Column	Meaning
source	natural or denovo
metric_name / canonical_key / input_class	the quantity, the distinct metric it belongs to and its class
ame_flavour	crystal = active-site accuracy against the deposited structure; self_design = against the design's own model (never pooled)

Column	Meaning
in_main_set	1 for the four main-set metrics (each appears several times: mean, maximum and spread over the five predicted models are separate rows)
condition	same_ec, diff_ec, decoy or apo
n_systems	systems with a value (59 natural, 30 de novo)
mean_cognate / mean_condition / mean_delta	mean value under the cognate ligand, under the condition, and the paired difference
median_delta_over_sigma / frac_beyond_1sigma	median change in units of the seed-to-seed standard deviation, and the share of systems that move by more than one
auroc / auroc_lo / auroc_hi	direction-free AUROC of cognate against the condition, 95% interval over systems

Excerpt (first 3 rows where source = natural, condition = diff_ec, in_main_set = 1).

source	metric_name	condition	n_systems	mean_cognate	mean_condition	auroc
natural	ame_crystal_clash_max	diff_ec	58	3.2826	3.6103	0.6564
natural	ame_crystal_rmsd_max	diff_ec	59	1.4469	1.5572	0.5395
natural	ame_crystal_rmsd_mean	diff_ec	59	1.2516	1.2895	0.5306

Notes.

- n is 59 natural enzymes here and 55 in Table 7, whose evaluation set drops four enzymes whose ligand could not be built.
- Per-metric results of this table are descriptive; the ranking and the best-of-30 threshold are in Table 7.

Table S8. Substrate swap stratified by ligand size and charge

File. supplementary/S8_substrate_swap_size_strata.csv · 684 rows × 27 columns · public

Supports. Results 3.2; Methods 2.3 (size check, Eq. 10). **Cited in.** Methods 2.3; Results 3.2.

A row is one quantity, source, wrong-ligand condition and stratum of the difference in ligand size or charge, for one analysis variant.

The check of whether ligand size or charge explains the discrimination. For each headline quantity (ipTM, per-chain pTM minimum, the combined score, ligand clearance) and for the AME RMSD and the AME pass bit, the cells (an enzyme and one wrong ligand) are stratified by the difference in heavy-atom count (smaller, matched, larger), by the charge difference (same, different) and jointly. For each stratum it gives the paired win fraction, the AUROC of the absolute value, the Spearman correlation of the size difference with the change, intervals over systems and clusters, and whether the effect survives.

Key columns.

Column	Meaning
source / variant	natural or denovo; primary or the variant that excludes ipTM = 0 cells (excl_iptm0)
metric_name / role	the quantity and headline or supplementary
stratum_family / stratum	all, size (smaller, matched, larger), charge (q_same, q_diff) or joint, and the stratum
condition	same_ec, diff_ec, decoy or the three pooled (pooled_3)
k_cells / N_cells / k_systems / N_systems	cells and systems in the stratum
win_frac / win_lo95 / win_hi95	paired win fraction (cognate higher than wrong ligand) with 95% interval

Column	Meaning
auroc_abs / auroc_abs_lo95 / auroc_abs_hi95 / survives	AUROC of the absolute value with 95% interval, and 1 / 0 / 'n<10, not estimated'
spearman_dn_vs_change / spearman_lo95 / spearman_hi95	Spearman correlation of the size difference with the change, with 95% interval
direction_declared / win_abs	the direction of change that counts as a win, and the win fraction of the absolute value
k_clusters / N_clusters / win_lo95_cluster / win_hi95_cluster	clusters and the cluster-level interval of the win fraction (where estimated)

Excerpt (first 3 rows where variant = primary, source = natural, stratum_family = size).

metric_name	stratum	condition	k_cells	N_cells	win_frac
chai_iptm_mean	smaller	same_ec	18	177	1.0
chai_aggregate_score_mean	smaller	same_ec	18	177	1.0
chai_per_chain_ptm_min_mean	smaller	same_ec	18	177	0.8333

Notes.

- Released as a CSV only for the full 684 rows; the excerpt shows the structure.

Table S9. BglB: every analysed quantity ranked by rank correlation with the measured impairment (complete Table 8)

File. supplementary/S9_bglb_all_quantities_ranked.csv · 54 rows × 30 columns · D2DCure aggregate (licence of the source data unstated; aggregates only, no per-variant rows)

Supports. Results 3.3.1; the complete version of Table 8. **Cited in.** Methods 2.4, 2.4.1; Results 3.3.1.

A row is one quantity analysed on the 432 BglB variants (54 in all).

Every quantity analysed on the 432-variant evaluation set: 39 structure-space metrics and 15 prediction-based metrics, ranked by the Spearman correlation of the size of its change with the measured impairment, with 95% interval over the 175 positions. It carries the best-of-114 threshold, the correlation of the signed change, and beside them the AUROC of the binary contrast (impaired against wild-type-like variants) with its interval and threshold. 32 of the 54 exceed the best-of-114 threshold for the rank correlation.

Key columns.

Column	Meaning
rank / rank_in_table_8 / in_table_8	rank among 54, rank among the 32 items of Table 8 and membership of Table 8
metric_name / canonical_key / source / input_class / role	the quantity, its source (structure-space or prediction-based), class and role
n_variants	variants with a value (418 to 432)
rho / rho_ci_lo / rho_ci_hi	Spearman correlation and 95% interval over positions
above_best_of_all_quantities_threshold	1 if the correlation is above the best-of-114 threshold
rho_signed_delta	correlation of the signed change

Column	Meaning
auroc / auroc_ci_lo / auroc_ci_hi	AUROC of the binary contrast (impaired against wild-type-like variants), computed on the size of the change and so not direction-free, with 95% interval; the column that follows is the threshold flag for the AUROC

The shared block of 11 provenance columns (see Conventions) is also present and is not repeated above.

Excerpt (first 3 rows).

rank	metric_name	role	rho	rho_ci_lo	rho_ci_hi
1	propka_catalytic_pka_range	main	0.48	0.3474	0.592
2	propka_catalytic_pka_min	main	0.4622	0.3224	0.5748
3	propka_catalytic_shift_absmax	main	0.4587	0.319	0.5782

Notes.

- Aggregates of D2DCure data (licence unstated); no per-variant rows.

Table S10. BglB: all 39 structure-space metrics

File. supplementary/S10_bglb_all_metrics.csv · 39 rows × 26 columns · D2DCure aggregate (licence of the source data unstated; aggregates only, no per-variant rows)

Supports. Results 3.3.1; Methods 2.4.1. **Cited in.** Methods 2.4.1; Results 3.3.1.

A row is one metric analysed on the 432 BglB variants (39 in all).

Spearman correlation, and signed, direction-free and magnitude AUROC with intervals, for every structure-space metric (36 site-scoped and 3 bookkeeping).

Key columns.

Column	Meaning
metric / canonical_key / input_class / in_main_set	the metric, its class and main-set membership
rho / rho_lo / rho_hi	Spearman correlation of the signed change of the metric with the impairment, with 95% interval over positions
auc_signed / auc_dirfree / auc_mag (+ _lo, _hi)	AUROC of the binary contrast using the signed change, the direction-free form and the magnitude, with intervals

The shared block of 11 provenance columns (see Conventions) is also present and is not repeated above.

Excerpt (first 3 rows where in_main_set = 1).

metric	in_main_set	rho	rho_lo	rho_hi	auc_dirfree
catalytic_frac_buried	1	-0.1909	-0.3048	-0.057	0.5312

metric	in_main_set	rho	rho_lo	rho_hi	auc_dirfree
catalytic_pairwise_dist_max	1	0.0431	-0.1841	0.2709	0.5149
catalytic_pairwise_dist_mean	1	0.09	-0.074	0.2468	0.5534

Notes.

- Aggregates of D2DCure data (licence unstated); no per-variant rows.

Table S11. BglB: prediction-based metrics (Chai-1 with the assay substrate)

File. `supplementary/S11_bglb_prediction_based_all_metrics.csv` · 15 rows × 26 columns · D2DCure aggregate (licence of the source data unstated; aggregates only, no per-variant rows)

Supports. Results 3.3.1; Methods 2.4.1. **Cited in.** Methods 2.4.1; Results 3.3.1.

A row is one prediction-based quantity analysed on the 432 BglB variants (15 in all).

The same columns as S10 for the quantities computed from Chai-1 predictions of each variant with the assay substrate (pNPG, no metal); active-site accuracy is measured against the wild-type crystal, which carries a covalent glucosyl intermediate. Here rho is the correlation of the signed change, as in S10. Twelve rows are main-set metrics; the largest signed correlation is 0.17 (AME RMSD, naive minimum) and the largest magnitude is 0.25 (ipTM and the Chai-1 combined score, a rescaled ipTM, both -0.25); the size-of-change correlation of Table 8 is in S9 (ipTM 0.33).

Key columns.

Column	Meaning
`metric` / `canonical_key` / `input_class` / `in_main_set`	the metric, its class and main-set membership
`rho` / `rho_lo` / `rho_hi`	Spearman correlation with 95% interval over positions
`auc_signed` / `auc_dirfree` / auc_mag (+ _lo, _hi)	AUROC of the binary contrast

The shared block of 11 provenance columns (see Conventions) is also present and is not repeated above.

Excerpt (first 3 rows where in_main_set = 1).

metric	in_main_set	rho	rho_lo	rho_hi
ame_crystal_clash_max	1	0.0828	-0.0534	0.2162
ame_crystal_rmsd_max	1	0.0632	-0.0622	0.1849
ame_crystal_rmsd_mean	1	0.1213	-0.0079	0.259

Notes.

- Aggregates of D2DCure data (licence unstated); no per-variant rows.

Table S12. Hits per 96-well plate for every metric on the 192 designs (filter-style analysis, active against inactive)

File. `supplementary/S12_hits_per_plate_192_designs.csv` · 71 rows × 36 columns · public

Supports. Results 3.3.2; Methods 2.4.2 (Eqs. 14 and 15). **Cited in.** Methods 2.4.2; Results 3.3.2.

A row is one metric scored on the 192 designs (71 in all).

The filter-style analysis, for every metric: the expected hits per 96-well plate and the actives among the 48 designs kept when the quarter of the 192 designs with the highest, or with the lowest, values of the metric fills the plate (8.0 hits when unfiltered), each with a 90% interval over designs; the metrics are ordered by the better end. The analysis uses the labels active and inactive and does not rank the designs by activity. 12 metrics exceed the random-filter band (14.0 hits).

Key columns.

Column	Meaning
`rank` / `rank_among_ranked_items` / `ranked_item`	rank among 71, rank among the 31 items of Table 9 and whether the metric is one of them
`metric_name` / `canonical_key` / `source` / `input_class` / `role`	the metric and its class
`n_designs` / `n_active` / `n_kept_highest` / `n_kept_lowest`	192, 16 and the number kept (fewer than 48 when designs tie at the cut)
`hits_highest_quarter` / `hits_highest_ci_lo90` / `hits_highest_ci_hi90` / `actives_in_highest_quarter`	hits per plate for the highest quarter, 90% interval and actives kept
`hits_lowest_quarter` / `hits_lowest_ci_lo90` / `hits_lowest_ci_hi90` / `actives_in_lowest_quarter`	the same for the lowest quarter
`better_end` / `best_hits_per_plate` / `above_random_band`	the better end, its hits per plate and whether it exceeds the random band
`note`	for example a tie at the cut

The shared block of 11 provenance columns (see Conventions) is also present and is not repeated above.

Excerpt (first 3 rows).

rank	metric_name	role	better_end	best_hits_per_plate	above_random_band
1	rosetta_cat_fa_intra_rep	main	lowest	22.0	1
2	rosetta_cat_fa_rep	main	lowest	22.0	1
3	rosetta_catalytic_vdw_rep	main	lowest	22.0	1

Notes.

- Descriptive only: no hypothesis test.

Table S13. Every metric ranked by its correlation with kcat/KM over the 16 designs that have a value (complete Table 9)

File. `supplementary/S13_all_metrics_ranking_16_designs.csv` · 52 rows × 26 columns · public

Supports. Results 3.3.2; the complete version of Table 9. **Cited in.** Methods 2.4.2; Results 3.3.2.

A row is one metric with a rank correlation on the 16 designs with a reported kcat/KM (52 in all).

Every metric that has a correlation, ranked by the size of the Spearman correlation with log10 kcat/KM, with a 90% interval over designs, whether the interval excludes 0 (12 do) and the direction.

Key columns.

Column	Meaning
`rank` / `rank_in_table_9` / `in_table_9`	rank among 52, rank among the 31 items of Table 9 and membership of Table 9
`metric_name` / `canonical_key` / `source` / `input_class` / `role`	the metric and its class
`n_designs` / `rho` / `rho_ci_lo90` / `rho_ci_hi90`	16 designs, Spearman correlation and 90% interval over designs
`interval_excludes_zero` / `higher_value_means`	whether the interval excludes 0 and whether a larger value means higher or lower activity

The shared block of 11 provenance columns (see Conventions) is also present and is not repeated above.

Excerpt (first 3 rows).

rank	metric_name	rho	rho_ci_lo90	rho_ci_hi90	higher_value_means
1	ame_self_rmsd_min	-0.6843	-0.9359	-0.2752	lower activity
2	ame_self_rmsd_naive_min	-0.6843	-0.9359	-0.2752	lower activity
3	ame_self_rmsd_mean	-0.6755	-0.946	-0.2801	lower activity

Notes.

- Descriptive analysis on the 16 designs that have a reported kcat/KM (8 backbone clusters); the other 176 of the 192 designs have no measured activity and are not ranked.

Table S14. Enrichment on the 192 designs at every keep fraction, structure-space metrics (active against inactive)

File. `supplementary/S14_enrichment_192_designs.csv` · 440 rows × 26 columns · public

Supports. Results 3.3.2; Methods 2.4.2. **Cited in.** Methods 2.4.2; Results 3.3.2.

A row is one metric, one direction (keep the highest or the lowest values) and one keep fraction (440 rows).

Enrichment on the 192 designs, from the labels active and inactive (the designs are not ranked by activity), for the 55 metrics of the structure-space panel, in both directions and at keep fractions of 5, 10, 25 and 50%: the number kept, the actives kept and the expected hits per 96-well plate with a 90% interval against the unfiltered 8.0.

Key columns.

Column	Meaning
metric / canonical_key / input_class / in_main_set	the metric
direction / keep_fraction / n_kept	keep the highest or the lowest values; the fraction and the number of designs kept
n_active_kept / n_active_total	actives among those kept, and 16
expected_hits_per_plate / hits_lo90 / hits_hi90	expected hits per plate with 90% interval
unfiltered_hits_per_plate / enrichment_vs_unfiltered	8.0 and the ratio to it

The shared block of 11 provenance columns (see Conventions) is also present and is not repeated above.

Excerpt (first 3 rows where in_main_set = 1, keep_fraction = 0.25).

metric	direction	keep_fraction	n_kept	n_active_kept	expected_hits_per_plate
catalytic_frac_buried	low	0.25	48	2	4.0
catalytic_frac_buried	high	0.25	48	1	2.0
catalytic_pairwise_dist_max	low	0.25	48	3	6.0

Notes.

- Descriptive only. Released as a CSV only for the full 440 rows.

Table S15. Enrichment on the 192 designs, prediction-based metrics (active against inactive)

File. supplementary/S15_prediction_based_enrichment_192_designs.csv · 128 rows × 26 columns · public

Supports. Results 3.3.2; Methods 2.4.2. **Cited in.** Methods 2.4.2; Results 3.3.2.

A row is one prediction-based metric, one direction and one keep fraction (128 rows).

The same as S14 for the 16 prediction-based metrics (Chai-1 with the transition-state analogue and zinc; active-site accuracy against the design's own model).

Key columns.

Column	Meaning
metric / canonical_key / input_class / in_main_set	the metric
direction / keep_fraction / n_kept / n_active_kept	as in S14
expected_hits_per_plate / hits_lo90 / hits_hi90 / enrichment_vs_unfiltered	expected hits per plate, 90% interval and ratio to 8.0

The shared block of 11 provenance columns (see Conventions) is also present and is not repeated above.

Excerpt (first 3 rows where keep_fraction = 0.25, direction = high).

metric	direction	keep_fraction	n_active_kept	expected_hits_per_plate
ame_self_clash_max	high	0.25	5	10.0
ame_self_pass	high	0.25	2	4.0
ame_self_rmsd_max	high	0.25	3	6.0

Notes.

- Descriptive only (16 active designs).

Table S16. Sanity floors by class of metric

File. supplementary/S16_sanity_floor_by_input_class.csv · 19 rows × 8 columns · public

Supports. Methods 2.5 (detection floor); Results 3.4.1. **Cited in.** Methods 2.5; Results 3.4.1.

A row is one sanity-floor condition on one data set and one class of metric (19 rows).

The sanity-floor conditions (all catalytic residues replaced by Gly, scrambled sequence on the native backbone, ligand removed on the 53-enzyme set) for the 143-enzyme set, the shortened-motif set, the 53-enzyme set and the de novo set: how many metrics of each class were scored and how many responded. One row reconciles the detection floor with the Gly step: the responses are identical for every scored metric (the same change scored twice).

Key columns.

Column	Meaning
dataset / condition / input_class	the data set, the condition and the class of metric
n_metrics / n_scored / n_responds / n_unscored	metrics in the class, scored, responding and not scored
note	remarks, including the reconciliation

Excerpt (first 2 rows where dataset = main set (143), condition = cat_to_gly).

dataset	condition	input_class	n_metrics	n_scored	n_responds
main set (143)	cat_to_gly	bookkeeping	20	20	8
main set (143)	cat_to_gly	struct_site	34	26	25

Table S17. Control matching: the loosest tier needed

File. supplementary/S17_control_match_tiers.csv · 45 rows × 6 columns · public

Supports. Methods 2.5; Results 3.4.1. **Cited in.** Methods 2.5; Results 3.4.1.

A row is one analysis, lesion test and matching tier (45 rows).

For each analysis and lesion test, how many (enzyme, step) entries needed each loosest matching tier: 0 = exact match, tiers 3 and above drop packing matching, tier 5 also relaxes burial (main set only), -1 = no control found; and the share of the lesion test.

Key columns.

Column	Meaning
analysis_set / lesion_test	the analysis (data set and controls) and the lesion test (catalytic lesion or second-shell lesion)
worst_tier / n_entries / share_of_lesion_test	loosest tier used, number of entries and their share of the lesion test
note	definition of the tiers

Excerpt (first 3 rows where analysis_set = main set, packing-matched controls, lesion_test = catalytic lesion).

analysis_set	lesion_test	worst_tier	n_entries	share_of_lesion_test
main set, packing-matched controls	catalytic lesion	-1.0	8	0.0035
main set, packing-matched controls	catalytic lesion	0.0	226	0.0997
main set, packing-matched controls	catalytic lesion	2.0	496	0.2189

Table S18. Control matching: balance of residues changed in the two arms

File. supplementary/S18_control_arm_balance.csv · 10 rows × 9 columns · public

Supports. Methods 2.5; Results 3.4.1. **Cited in.** Methods 2.5; Results 3.4.1.

A row is one analysis and lesion test (10 rows).

The share of (enzyme, step) pairs in which the control arm changed fewer or more residues than the catalytic arm, the median and mean ratio, and the number with no control. The comparison of medians used elsewhere cannot see this imbalance.

Key columns.

Column	Meaning
analysis_set / lesion_test / pairs	the analysis, the lesion test and the number of pairs
control_fewer_share / control_more_share	shares with fewer or more residues changed in the control arm
median_ratio / mean_ratio / catalytic_only_no_control	ratio of residues changed and pairs with no control

Excerpt (first 3 rows).

analysis_set	lesion_test	pairs	control_fewer_share	control_more_share	median_ratio
53-enzyme set	catalytic lesion	212	0.198113	0.0	1.0
53-enzyme set	second-shell lesion	159	0.0	0.0	1.0
main set	catalytic lesion	563	0.246892	0.0	1.0

Table S19. Catalytic lesion: every metric scored (Table 10 plus supplementary metrics and per-step flags)

File. `supplementary/S19_catalytic_lesion_all_metrics_ranked.csv` · 41 rows × 65 columns · public

Supports. Results 3.4.1; the complete version of Table 10. **Cited in.** Methods 2.5.1; Results 3.4.1.

A row is one metric scored on the catalytic-lesion ladder (41 rows, 39 distinct metrics).

Table 10 with 8 further rows (6 supplementary metrics and 2 aliases of ranked metrics) and every flag that decides whether a step counts. Metrics are grouped by the lesion step at which they are first detected and ranked by the mean of the rank statistic R over the counted steps. For each step (isosteric 15.3, non-isosteric 36.4, Ala 53.2, Gly 80.5 A3) it gives R with interval, the number of informative proteins and the flags that the metric responds, that the control response is above its noise and that the metric is specific; the number of steps counted; the first step detected and the first step at which the metric is specific (the onset of specificity); and the mean R of the 143 main enzymes alone. Structure-space metrics: 195 pooled natural enzymes (143 + 52 from the 53-enzyme set); prediction-based: 59 enzymes, 286 pairs.

Key columns.

Column	Meaning
`detection_group` / `rank_in_group` / `rank` / `rank_in_table_10`	group by first detected step, rank in the group, overall rank and rank among the 33 items of Table 10
`metric_name` / `canonical_key` / `source` / `input_class` / `role` / `in_ranked_set`	the metric, class, role and whether it is one of the 33 ranked items
`n_steps` / `n_steps_interpretable`	steps with a value and steps counted
`r_mean` / `r_lo` / `r_hi`	mean R over counted steps with 95% interval over enzyme sub-subclasses
r_ / _lo / *hi* / *n*	R, interval and informative proteins for isosteric, non_isosteric, ala and gly
responds_ / control_above_ noise_ / specific_	the per-step flags
`first_step_detected` / `first_step_specific`	onsets of detection and of specificity
`r_mean_primary_only` / `above_best_of_all_quantities_threshold`	mean R of the 143 main enzymes and 1 if the mean R exceeds the best-of-33 threshold of Table 10
`r_mean_all_steps` / `r_all_lo` / `r_all_hi`	mean R over all steps with a value (counted or not) with 95% interval
`r_primary_lo` / `r_primary_hi` / `r_mean_all_steps_primary_only`	95% interval of the mean R of the 143 main enzymes, and its mean over all steps with a value
`applicability_status`	applicability status

The shared block of 11 provenance columns (see Conventions) is also present and is not repeated above.

Excerpt (first 3 rows).

detection_group	rank_in_group	metric_name	r_mean	r_lo	r_hi
isosteric	1	chai_per_chain_ptm_min_mean	0.7524	0.6768	0.8238
isosteric	2	propka_catalytic_pka_range	0.6834	0.5759	0.7806
isosteric	3	chai_iptm_mean	0.6732	0.5991	0.7463

Notes.

- S20 holds the counts of the specificity-ratio analysis, S23 the ratios.

Table S20. Structure-space specificity by metric family and lesion step, with the panel statistic and sensitivity sets

File. `supplementary/S20_specificity_by_family_and_step.csv` · 20 rows × 26 columns · public

Supports. Results 3.4.1 and 3.4.2. **Cited in.** Methods 2.5.1; Results 3.4.1, 3.4.2.

A row is one family of metrics at one lesion step or dose (20 rows).

The number of eligible, responding and specific structure-space metrics (as distinct metrics) by family (site-scoped Rosetta energy, catalytic geometry, catalytic pKa, all main-set metrics) at each step of the catalytic lesion and each second-shell dose, the geometric-mean specificity ratio of the main-set panel (isosteric, 1.075 [0.958, 1.178]), and the counts in four sensitivity sets (shortened motif, no packing matching or count gate, the 53-enzyme set, de novo designs). At the isosteric step 23 of 29 metrics respond and none is specific; specificity appears for 5 metrics at the non-isosteric step and for 8 at the Ala and Gly steps.

Key columns.

Column	Meaning
`family` / `lesion_test` / `level` / `dose_value` / `dose_unit`	family, lesion test (catalytic lesion or second-shell lesion), step or dose and its size
`n_eligible_keys` / `n_responds_keys` / `n_specific_keys` / `n_cells`	counts as distinct metrics, and cells
`sr_geomean_ci90`	geometric-mean specificity ratio of the panel with 90% interval
`sens_shortened_motif` / `sens_no_packing_match` / `sens_53_enzyme_set` / `sens_de_novo_set`	specific / eligible in the shortened-motif, no-packing-match, 53-enzyme and de novo sets

The shared block of 11 provenance columns (see Conventions) is also present and is not repeated above.

Excerpt (first 3 rows where lesion_test = catalytic lesion).

family	level	n_eligible_keys	n_responds_keys	n_specific_keys	sens_shortened_motif
Rosetta energy (site-scoped)	isosteric	15	13	0	0 of 15
Catalytic geometry (site-scoped)	isosteric	8	5	0	0 of 8
Catalytic pKa	isosteric	6	5	0	

Table S21. Sensitivity of the specificity-ratio counts to motif size, packing matching and count gating

File. `supplementary/S21_motif_packing_sensitivity.csv` · 20 rows × 7 columns · public

Supports. Results 3.4.1. **Cited in.** Methods 2.5.1; Results 3.4.1.

A row is one analysis and one lesion step (20 rows).

Counts of eligible, responding and specific metrics at each step for five analyses: the 53-enzyme set, the main set without packing matching or count gating, the main set with packing-matched controls (primary), the main set with the motif shortened to at most 3 residues, and the 30 de novo designs with a created-site control.

Key columns.

Column	Meaning
`reader_name` / `motif_and_controls`	the analysis and how its motif and controls are defined
`level`	lesion step
`n_eligible_keys` / `n_responds_keys` / `n_specific_keys` / `n_cells`	counts as distinct metrics, and cells

Excerpt (first 3 rows).

reader_name	level	n_eligible_keys	n_responds_keys	n_specific_keys
53-enzyme set	isosteric	23	12	0
53-enzyme set	non_isosteric	23	12	0
53-enzyme set	ala	23	14	0

Table S22. Re-predicted ladder: seed-noise context

File. `supplementary/S22_replicate_noise_context.csv` · 40 rows × 12 columns · public

Supports. Methods 2.5.1; Results 3.4.1. **Cited in.** Methods 2.5.1; Results 3.4.1.

A row is one prediction-based metric in one arm and subset (40 rows).

Noise context from the second-seed replicate: the median absolute change in each arm against the seed noise of the metric, for the seven prediction-based metrics, on all pairs and on the original matched subset.

Key columns.

Column	Meaning
`source` / `metric_name` / `subset` / `arm`	natural or de novo, the metric, all_pairs or original_matched, and catalytic or control
`k_pairs` / `N_pairs` / `k_systems` / `N_systems`	pairs and systems
`median_abs_delta` / `median_sigma_native` / `frac_abs_delta_gt_sigma_native`	median absolute change, the metric's seed noise and the share of pairs whose change exceeds it

Excerpt (first 3 rows where source = natural, subset = all_pairs).

metric_name	subset	arm	median_abs_delta	median_sigma_native	frac_abs_delta_gt_sigma_native
chai_iptm_mean	all_pairs	catalytic	0.00666	0.00154	0.8292
chai_iptm_mean	all_pairs	control	0.00254	0.00154	0.6797
chai_aggregate_score_mean	all_pairs	catalytic	0.00556	0.00127	0.8392

Table S23. Specificity ratio of every eligible metric at every step

File. `supplementary/S23_eligible_metrics.csv` · 256 rows × 28 columns · public

Supports. Methods 2.5.1 (Eq. 20); Results 3.4.1 and 3.4.2. **Cited in.** Methods 2.5.1; Results 3.4.1, 3.4.2.

A row is one eligible (site-scoped) metric at one lesion step or dose (256 rows).

The median change at the catalytic lesion and its interval, the specificity ratio with 95% interval, whether the metric responds and whether it is specific, and the verdict: specific, non_specific, blind (does not respond), invariant (does not change) or below the floor of five units. Steps of the catalytic lesion are isosteric, non_isosteric, ala, gly; second-shell doses are second_shell_1, _2, _4.

Key columns.

Column	Meaning
`canonical_key` / `metric_name` / `input_class` / `in_main_set`	the metric (aliases collapse in canonical_key)
`lesion_test` / `level` / `n_systems`	catalytic lesion or second-shell lesion, the step or dose and the number of enzymes
`delta_median` / `delta_ci_lo` / `delta_ci_hi`	median change at the catalytic lesion with interval
`sr_median` / `sr_ci_lo` / `sr_ci_hi`	specificity ratio with 95% interval
`responds` / `specific` / `verdict` / `applicability_status`	the flags, the verdict and the applicability status

The shared block of 11 provenance columns (see Conventions) is also present and is not repeated above.

Excerpt (first 3 rows where in_main_set = 1, level = ala).

canonical_key	lesion_test	level	n_systems	sr_median	verdict
catalytic_frac_buried	catalytic lesion	ala	143	1.9091	specific
catalytic_pairwise_dist_max	catalytic lesion	ala	124	0.0	blind
catalytic_pairwise_dist_mean	catalytic lesion	ala	124	0.7262	non_specific

Table S24. The specific cells that survive the controls

File. `supplementary/S24_survivors_by_status.csv` · 70 rows × 10 columns · public

Supports. Methods 2.5.1; Results 3.4.1. **Cited in.** Methods 2.5.1; Results 3.4.1.

A row is one metric at one step in one analysis (70 rows: 50 natural, 20 de novo).

The cells that are specific under the control of their analysis, with the specificity ratio, the population (natural or de novo) and the applicability status of the metric. Of the 70, 69 are of metrics with status OK and 1 of a metric with too little coverage.

Key columns.

Column	Meaning
analysis_set / lesion_test / level	the analysis, the lesion test and the step
metric_name / canonical_key / input_class / in_main_set	the metric
sr_text	specificity ratio with interval, as text
applicability_status / population	applicability status and natural or de novo

Excerpt (first 3 rows where analysis_set = main set, packing-matched controls, in_main_set = 1).

analysis_set	lesion_test	level	metric_name	sr_text	population
main set, packing-matched controls	catalytic lesion	ala	catalytic_ frac_buried	1.91 [1.4, 2.72]	natural
main set, packing-matched controls	catalytic lesion	ala	catalytic_ radius_of_ gyration	1.76 [1.43, 2.19]	natural
main set, packing-matched controls	catalytic lesion	ala	catalytic_rel_ sasa_mean	2.4 [1.93, 3.03]	natural

Table S25. Rank statistic R per lesion step on the 143 main enzymes

File. supplementary/S25_ranking_statistic_by_lesion_step.csv · 151 rows × 14 columns · public

Supports. Methods 2.5.1 (Eq. 17). **Cited in.** Methods 2.5.1; Results 3.4.1.

A row is one metric at one step of the catalytic lesion (151 rows).

The rank statistic R (probability that a metric changes more at the catalytic lesion than at the matched control) for the 143 main enzymes alone, with 90% intervals, for 42 distinct metrics. This is the per-step form of R on the main enzymes; Table 10 and S19 report the pooled analysis on 195 enzymes with 95% intervals and the counted-step rule, so the values for the same metric differ.

Key columns.

Column	Meaning
metric_name / canonical_key / input_class / in_main_set	the metric
lesion_step / volume_A3	lesion step and its median side-chain volume change
R / R_lo90 / R_hi90 / n_systems / n_clusters	R with 90% interval, enzymes and enzyme sub-subclasses
responds / interpretable	whether the metric responds and whether R is interpretable
applicability_status	applicability status

Excerpt (first 3 rows where in_main_set = 1, lesion_step = ala).

metric_name	lesion_step	R	R_lo90	R_hi90	n_systems
catalytic_ frac_buried	ala	0.7557	0.686	0.8189	131
catalytic_ pairwise_dist_ max	ala	0.203	0.1522	0.2574	133

metric_name	lesion_step	R	R_lo90	R_hi90	n_systems
catalytic_pairwise_dist_mean	ala	0.3262	0.2553	0.4	141

Table S26. Panel equivalence statistic for the main-set metrics

File. supplementary/S26_panel_equivalence_main_set.csv · 2 rows × 10 columns · public

Supports. Results 3.4.1; Methods 2.5.1 (equivalence to 1). **Cited in.** Methods 2.5.1; Results 3.4.1.

A row is one analysis and set of metrics (2 rows).

The geometric-mean specificity ratio of the panel with a 90% interval (nested bootstrap that recomputes every metric inside each draw), for the 27 main-set metrics with a defined ratio (isosteric step: 1.075 [0.958, 1.178] with packing-matched controls, 1.022 [0.909, 1.125] without packing matching).

Key columns.

Column	Meaning
analysis_set / metric_set / n_metrics	the analysis, the set of metrics and its size
geomean_sr / ci90_lo / ci90_hi / median_sr	geometric-mean ratio, 90% interval and median
n_clusters / status / note	resampling clusters, status and remarks

Excerpt (first 2 rows).

analysis_set	metric_set	n_metrics	geomean_sr	ci90_lo	ci90_hi
main set, packing-matched controls	main-set metrics only	27	1.0746	0.9584	1.1782
main set	main-set metrics only	27	1.0215	0.9087	1.1249

Table S27. Re-predicted ladder: specificity ratio per step for all metrics

File. supplementary/S27_ladder_re_predicted_per_level.csv · 140 rows × 15 columns · public

Supports. Methods 2.5.1; Results 3.4.1. **Cited in.** Methods 2.5.1; Results 3.4.1.

A row is one metric at one step for one source (140 rows).

The re-predicted catalytic-lesion ladder per step (ALL steps pooled, isosteric, non-isosteric, Ala, Gly) for the 31 named quantities (17 distinct metrics), natural and de novo: pairs and systems and the specificity ratio (median and mean) with intervals.

Key columns.

Column	Meaning
source / metric_name / canonical_key / input_class / ame_flavour	the metric
level / n_pairs / n_systems	step and size

Column	Meaning
sr_median / sr_lo / sr_hi	ratio of medians with interval
sr_mean / sr_mean_lo / sr_mean_hi	ratio of means with interval
applicability_status	applicability status

Excerpt (first 3 rows where source = natural, level = isosteric, canonical_key = chai_iptm).

metric_name	level	n_pairs	n_systems	sr_mean	sr_mean_lo	sr_mean_hi
chai_iptm_max	isosteric	118	59	1.622	1.058	2.526
chai_iptm_mean	isosteric	118	59	1.963	1.44	2.707
chai_iptm_std	isosteric	118	59	1.046	0.605	1.878

Table S28. Second-shell lesion: every metric scored, with R at each dose and the per-dose flags

File. supplementary/S28_second_shell_lesion_all_metrics_ranked.csv · 37 rows × 58 columns · public

Supports. Results 3.4.2; Methods 2.5.2. **Cited in.** Methods 2.5.2; Results 3.4, 3.4.2.

A row is one metric scored on second-shell removal (37 rows, 35 distinct metrics).

The 29 ranked metrics and 8 further rows (6 supplementary metrics and 2 aliases of ranked metrics), with the flags at each dose (1, 2 and 4 residues removed): R with interval, the number of informative proteins, and whether the metric responds, whether the control response is above its noise and whether it is specific. No dose counts for any metric, so the raw values are reported and no reference level is computed.

Key columns.

Column	Meaning
detection_group / rank_in_group / rank / rank_among_ranked_items	group by first dose detected, rank in the group, overall rank and rank among the 29 ranked metrics
metric_name / canonical_key / source / input_class / role / in_ranked_set	the metric and its role
n_steps / n_steps_interpretable	doses with a value and counted (0 for every metric)
r_second_shell_ / _lo / *hi* / *n_second_shell*	R, interval and informative proteins at dose 1, 2 and 4
responds_second_shell_ / control_above_noise_second_shell_ / specific_second_shell_	per-dose flags
r_mean_all_steps / first_step_detected / first_step_specific	mean R over all doses and the onsets
r_mean / r_lo / r_hi	mean R over the counted doses with interval (empty: no dose counts)
r_all_lo / r_all_hi / r_mean_primary_only / r_primary_lo / r_primary_hi / above_best_of_all_quantities_threshold	interval of the mean over all doses; the same for the 143 main enzymes alone; reference-level flag (empty: no reference level)

The shared block of 11 provenance columns (see Conventions) is also present and is not repeated above.

Excerpt (first 3 rows).

detection_group	rank_in_group	metric_name	r_mean_all_steps	n_steps_interpretable
second_shell_4	1	propka_catalytic_shift_mean	0.7005	0
second_shell_4	2	propka_catalytic_pka_mean	0.6915	0
never	1	catalytic_frac_buried	0.781	0

Notes.

- Both arms are scored over the catalytic residues.

Table S29. Effective dimensionality of the panel

File. supplementary/S29_effective_dimensionality.csv · 2 rows × 10 columns · public

Supports. Results 3.4.1. **Cited in.** Results 3.4.1.

A row is one set of metrics and one panel (2 rows).

The participation ratio and related measures of the effective number of independent metrics, for the 29 structure-space main-set metrics, for native values (7.85 of 29) and for isosteric responses (10.5 of 23).

Key columns.

Column	Meaning
metric_set / panel / n_metrics / n_systems_complete	the set, the panel and sizes
participation_ratio / effective_rank_fraction	effective number of independent metrics and its share
var_explained_pc1 / n_pcs_for_90pct / median_abs_offdiag_r	variance in the first component, components for 90% and median absolute correlation between metrics

Excerpt (first 2 rows).

metric_set	panel	n_metrics	participation_ratio	var_explained_pc1
main-set metrics (one member per ...	native values across systems	29	7.85	0.207
main-set metrics (one member per ...	isosteric responses (catalytic ar...	23	10.5	0.188

Table S30. De novo designs: burial equivalence of the created-site control

File. supplementary/S30_denovo_created_control.csv · 4 rows × 11 columns · public

Supports. Results 3.4.1. **Cited in.** Results 3.4.1.

A row is one burial metric (4 rows).

In the 30 de novo designs the control is a position mutated to the catalytic residue type. For the four burial metrics the table gives the catalytic-arm and control-arm medians, the paired difference with 90% interval and whether they differ.

Key columns.

Column	Meaning
`metric` / `n_systems` / `n_clusters`	the burial metric, designs and backbone clusters
`catalytic_arm_median` / `control_arm_median`	medians in the two arms
`paired_diff_median` / `paired_diff_ci90_lo` / `paired_diff_ci90_hi` / `ratio_median`	paired difference with interval and ratio
`differs` / `note`	1 if the arms differ

Excerpt (first 3 rows).

metric	n_systems	catalytic_arm_median	control_arm_median	paired_diff_median	differs
catalytic_rel_sasa_mean	30	0.143345	0.1315	0.027754	0
catalytic_rel_sasa_max	30	0.21568	0.204212	0.025605	0
catalytic_sasa_mean	30	32.109173	29.455957	6.216884	0

Table S31. PLACER ensemble metrics on the isosteric step (not interpretable)

File. `supplementary/S31_placer_isosteric.csv` · 50 rows × 12 columns · public

Supports. Results 3.4.1. **Cited in.** Results 3.4.1.

A row is one PLACER metric in one analysis (50 rows).

The specificity ratio, verdict and applicability of 25 PLACER ensemble metrics on the isosteric step, in two analyses. They are not interpretable: the control arm is covered 1.6 to 3.2 times less than the catalytic arm in every crop configuration, so the count of specific metrics moves with the crop anchor alone.

Key columns.

Column	Meaning
`analysis_set` / `metric_name` / `level` / `n_systems`	the analysis, the metric, the step and enzymes
`sr_median` / `sr_ci_lo` / `sr_ci_hi`	specificity ratio with 95% interval
`responds` / `specific` / `verdict` / `applicability_status`	flags, verdict (blind, non_specific, specific, no_dynamic_range) and status (GLOBAL or X-CONFOUND)

Excerpt (first 3 rows).

analysis_set	metric_name	n_systems	sr_median	verdict	applicability_status
main set, packing-matched controls	placer_catalytic_rmsf_max	132	1.0106	blind	X-CONFOUND
main set, packing-matched controls	placer_catalytic_rmsf_mean	132	0.9058	blind	X-CONFOUND
main set, packing-matched controls	placer_centroid_dev_mean	132	0.9237	blind	X-CONFOUND

Appendix A. Main-text tables

The six data tables of the Results (Tables 5 to 10), with the supplementary tables that hold the complete version or more detail. Tables 1 to 4 of the Methods are descriptive and are not released as files.

Table	File	Rows × columns	More detail in
T5	`main/T5_dead_vs_active.csv`	26 × 24	S5, S3
T6	`main/T6_plated_designs_auroc.csv`	25 × 26	S6
T7	`main/T7_substrate_swap.csv`	17 × 31	S7, S8
T8	`main/T8_bglb_variants.csv`	26 × 23	S9, S10, S11
T9	`main/T9_metric_ranking_16_designs.csv`	25 × 24	S13, S12, S14, S15
T10	`main/T10_specificity_catalytic_lesion.csv`	33 × 30	S19, S20, S23, S24, S25, S21

The second-shell lesion (Results 3.4.2) has no data table of its own: its values are in S28.

Columns of the data tables. Each data table has the shared provenance block (Conventions) and the columns of its complete table: `rank`, `item` or `metric` (display name and canonical key; `quantity` and `summary_over_models` in T7), `role`, `mode`, `family`, the number of units (`n_pairs`, `n_designs` and `n_active`, `n_systems`, `n_variants`), the estimate with its interval (`auroc` or `mean_auroc`, `rho`, `r_mean`, with `_ci_lo` and `_ci_hi` for 95% intervals and `_lo90` and `_hi90` for 90% intervals), `higher_in` or `higher_value_means` (the group in which the metric takes larger values), `auroc_directed`, `above_best_of_m_threshold` (1 if the estimate exceeds the best-of-M threshold of the table), `interval_excludes_zero` (T9) and `note`. T7 has one AUROC with its interval for each kind of wrong ligand (`same_ec`, `diff_ec`, `decoy`). T10 has `detection_group` and `rank_in_group` (rows are grouped by the lesion step at which the metric is first detected), `n_steps_counted`, `r_isosteric`, `r_non_isosteric`, `r_ala` and `r_gly` (R at each step, given also where the step is not counted) and `first_step_specific`.

Appendix B. Notes on the data

- Two sets of numbers for natural enzymes in the substrate swap: S7 uses all 59 natural enzymes, Table 7 uses the evaluation set of 55.
- Two forms of the rank statistic R: S25 is R on the 143 main enzymes with 90% intervals; Table 10 and S19 are the pooled analysis on 195 enzymes with 95% intervals and the counted-step rule. Values for the same metric differ.